CSMSS
Chhatrapati Shahu Maharaj Shikshan Sanstha's
CHH. SHAHU COLLEGE OF ENGINEERING
Kanchanwadi, Paithan Road, Chatrapati Sambhajinagar (Aurangabad)
DEPARTMENT OF ELECTRONICS & COMMUNICATION
(ADVANCED COMMUNICATION TECHNOLOGY)

COMMUNITY ENGAGEMENT PROJECT_S.Y._2026

Title : IoT-Based Weather Display Clock & Forecast Alert System

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Problem Statement Title : Low-cost smart clock that shows live weather and gives timely forecast alerts to communities

Theme : Smart Communities – IoT for Public Safety & Daily Life

PS Category- Software/Hardware : Hardware + Software (ESP8266 device with environmental sensors and alert software)

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- **Problem Statement**

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Problem Statement

Problem: Sudden rain, storms and heatwaves reach people without warning. Clocks show only time, phone apps need active checking, and weather stations are costly with no automatic alerts.

❖Proposed Solution (Describe your Idea/Solution/Prototype)

- **Detailed explanation of the proposed solution**

  ESP8266 smart clock showing time, humidity, pressure, rain status and air quality with a weather forecast on one OLED display, with buzzer alerts.

- **How it addresses the problem**

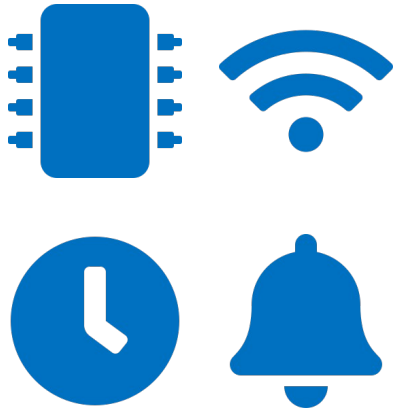
  Everything visible at a glance; automatic buzzer warnings for rain, poor air quality or a sudden pressure drop (storm); useful even for people who do not use smartphones.

- **Innovation and uniqueness of the solution**

  Clock + local sensing (humidity, pressure, rain, air quality) + forecast alerts in one affordable unit; RTC keeps time even when Wi-Fi is down.

1_Introduction

- IoT connects small devices to the internet to sense, share and act on data
- Project: ESP8266-based smart clock showing time, humidity, pressure, rain status and air quality
- Senses local conditions using DHT11, BMP280, YL-83 and MQ135, with an RTC module for timekeeping
- Buzzer warns users about rain, poor air quality or a falling-pressure (storm) condition
- Built with low-cost, easily available components for community use



2_Literature Survey

Sr.	Title / Research Work	Authors	Year	Main Technologies / Sensors Used	Key Features	Limitations
1	**IOT Based Weather Station**	N. Tenzin, J. Tharchen, K. Penjor, T. Dhendup, D. Lhamo	2021	NodeMCU (ESP8266), DHT11, LDR and other weather sensors	Measures temperature, humidity, pressure, light, rain and PM; logs data to ThingSpeak or a database	No timekeeping; no buzzer alert; not a compact clock-style device
2	**IoT-Based Weather Station Using an Embedded System**	O. Olanrele, O. A. Adeaga, O. A. Adeyemi, O. K. Ajayi, A. O. Mowemi	2022	NodeMCU ESP8266, DHT11, LDR	Monitors temperature, humidity, light intensity; uploads data to ThingSpeak for remote monitoring	No local display; no timekeeping; limited parameters
3	**Development of Blynk IoT Platform Weather Information Monitoring System**	M. A. Kamarudin, N. H. Mohd Yunus, M. R. Abd Razak, M. S. Mohd Nadzir, K. M. Alhasa	2023	ESP8266, DHT11, rain drop and CO sensors	Real-time weather monitoring; alerts via Blynk app, SMS and email	No compact display; no integrated clock
4	**Cloud-Based Weather Condition Monitoring System Using ESP8266 and Amazon Web Services**	A. Mohamed, G. Gunasegaran, D. Herath	2023	ESP8266, ultrasonic, raindrop and pressure sensors	Sends weather data over Wi-Fi to AWS IoT cloud; real-time reports on a website	No timekeeping; no local visual display; no buzzer alert

Gap: no single low-cost device combines clock + local sensing (pressure, humidity, rain, air quality) + buzzer alerts.

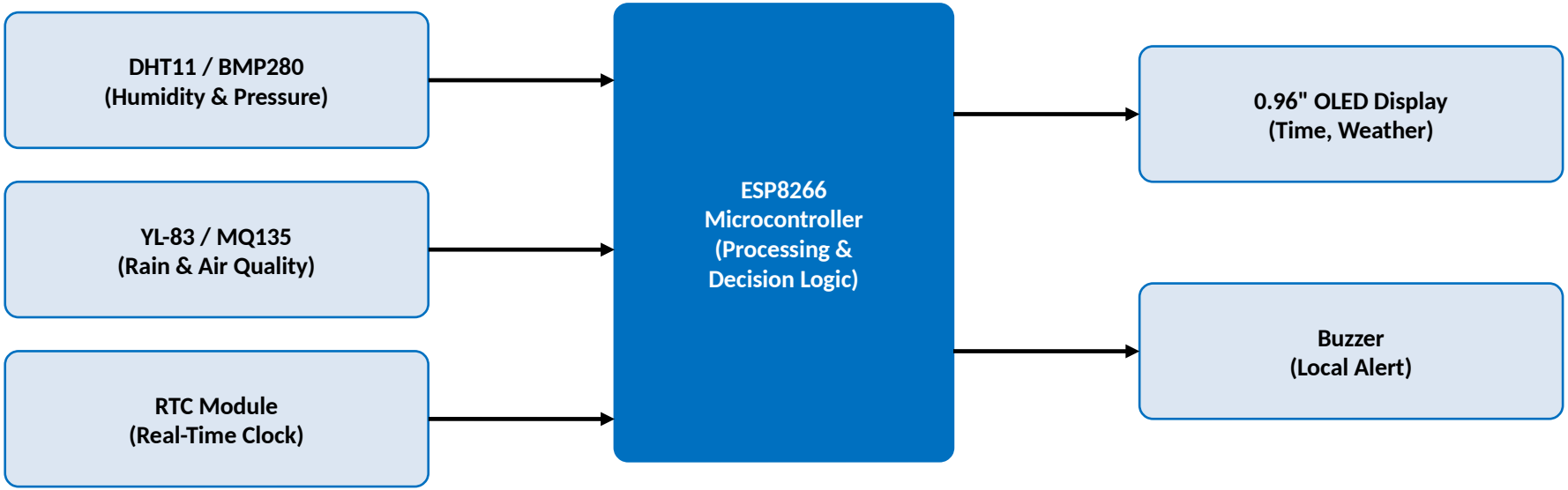
3_Proposed Solution

- ESP8266 clock with RTC for accurate time, even without Wi-Fi
- Humidity (DHT11), pressure (BMP280), rain (YL-83) and air quality (MQ135) on one 0.96" OLED
- Automatic alerts: buzzer + OLED message for rain, poor air quality or falling pressure
- Innovation: clock, local sensing and forecast alerts in one affordable unit
- Auto-syncs time over Wi-Fi (NTP); compact, low-power and easy to build



Concept view of the display

4_Working Operation/ Block Diagram



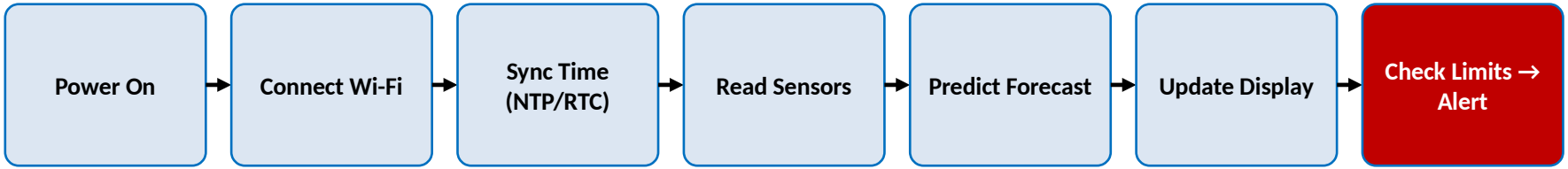
Working: ESP8266 reads the DHT11, BMP280, YL-83 and MQ135 sensor values and the RTC time, shows them on the 0.96" OLED display, compares values with set limits, and triggers the buzzer when limits are crossed.

5_Technnical Approach

- **Technologies to be used (e.g. programming languages, frameworks, hardware).**

  Hardware: ESP8266 (NodeMCU), RTC module, DHT11, BMP280, YL-83, MQ135, 0.96" OLED, buzzer, 5V supply. **Software:** Arduino IDE (C/C++), ESP8266 Wi-Fi library, DHT / BMP280 / OLED display libraries, RTC library, NTP time sync.

- **Methodology and process for implementation (Flow Charts/Images/ working prototype)**



Cycle repeats at regular intervals; RTC keeps time if Wi-Fi drops.

6_Feasibility And Viability

Analysis of the feasibility of the idea

- Low-cost, easily available parts
- Open-source Arduino tools and libraries
- Can be built within the semester

Potential challenges and risks

- Wi-Fi or power outage
- Wrong forecasts / sensor errors
- Sensor drift and noise

Strategies for overcoming these challenges

- RTC + last-known data offline
- Use pressure trend over time, not one reading
- Calibrate sensors (MQ135 warm-up); backup battery

7_Impact And Benefits

Potential impact on the target audience.

- Timely weather warnings for families, elders and workers
- Reduces surprise from rain, storms and heatwaves
- One glance gives time and weather

Benefits of the solution (social, economic, environmental, etc.)

- Social: safety and awareness
- Economic: affordable; helps plan work and reduce losses
- Environmental: low power use; air quality and weather awareness

8_Application



Homes



Schools & Colleges



Farms & Villages



Hospitals & Clinics



Shops & Offices



Public Spaces

9_Research And References

- **Details / Links of the reference and research work**

  - ESP8266 documentation – Espressif Systems: docs.espressif.com
  - BMP280 datasheet – Bosch Sensortec; DHT11 and MQ135 sensor datasheets
  - YL-83 rain sensor and RTC module (DS1307/DS3231) datasheets; 0.96" SSD1306 OLED datasheet
  - Arduino reference and ESP8266 Arduino core: docs.arduino.cc
  - Tenzin N. et al., "IOT Based Weather Station", 2021; Olanrele O. et al., "An IoT Based Weather Station Using an Embedded System", QJOEST, 2022
  - Kamarudin M. A. et al., "Development of Blynk IoT Platform Weather Information Monitoring System", 2023; Mohamed A. et al., "Cloud-based weather condition monitoring system using ESP8266 and Amazon Web Services", 2023

THANK YOU